YI LIN

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Summary

M.S. student in Computer Science at National Cheng Kung University, focusing on **Computer Vision** and **Deep Learning**. Experienced in applying **software engineering** principles to build modular and reusable AI systems. **First Place (Champion)** in the **2025 TSMC IT CareerHack**; published a **first-author** paper at **IEEE ICMAIL 2023**.

Education

- **National Cheng Kung University (NCKU)** — Tainan, Taiwan
  M.S. in Computer Science and Information Engineering — *Sep 2024 – Expected Jul 2026*
  - **Thesis**: Deep Learning-Based Pediatric Hip X-ray Analysis System
  - **Relevant Coursework**: Deep Learning, Computer Vision, Image Processing
- **National Taiwan Ocean University (NTOU)** — Keelung, Taiwan
  B.S. in Computer Science and Engineering (GPA: 3.77/4.0) — *Sep 2020 – Jun 2024*

Selected Projects

- **Deep Learning-Based Pediatric Hip X-ray Analysis System | Keypoint Detection** — Master's Thesis
  PyTorch, YOLO, ConvNeXt, SimCC, Medical Imaging — ***GitHub Code***
  - Built an end-to-end diagnostic pipeline for developmental dysplasia of the hip (DDH) using a **two-stage top-down** keypoint regression approach: **YOLO** for ROI localization and **ConvNeXt + SimCC** for sub-pixel keypoint regression.
  - Achieved **state-of-the-art** performance on the MTDDH dataset, improving over **YOLOv11-seg** from F1 0.593 to **0.943** (**+59%**).
  - Validated on clinical data with lower error than a senior physician (**Mean Error: 5.01px vs. 6.34px**), demonstrating practical value for **automated/assisted diagnosis**.
  - Designed a scalable **modular framework** supporting multiple backbones (HRNet, EfficientNet) and head variants for acetabular index (AI) angle measurement and IHDI classification.
- **AI Domain-based Translation System (2025 TSMC IT CareerHack)** — First Place (Champion)
  JavaScript, String Matching, Frontend Development — ***GitHub Code***
  - Developed a real-time translation tool for enterprise communication with automatic **domain-term highlighting**; built the frontend with **HTML/CSS/JavaScript**, winning **First Place**.
  - Implemented a custom **terminology matching** algorithm to auto-map and annotate technical terms in translations, improving readability and consistency.
  - Presented the winning product to TSMC executives, demonstrating strong technical communication and presentation skills.
- **Deep Learning-Based Underwater Image Color Restoration System** — Undergraduate Research
  PyTorch, GAN, Image Enhancement — ***Code*** ***Paper***
  - Published a conference paper as **First Author** at **IEEE ICMAIL 2023**.
  - Combined **WaterNet** (CNN-based underwater color restoration) with a **GAN**-based colorization model to mitigate red-light attenuation and color shift; evaluated results via palette analysis and quantitative metrics (**K-means + CIEDE2000**).

Experience

- **Programing.tw** — Taiwan
  Co-Founder & Product Lead — *Feb 2021 – Present*
  - **Full-Stack Development**: Built the "Proglearn" platform using **Go (Gin)** and **Vue.js**; implemented **WebSocket**-based real-time remote teaching.
  - **Awards**: Received **Top Award (Top 4)** at the Chao Chuang Ke Startup Competition, and **High Distinction** at the 2023 Information Intelligence Innovation Cross-Domain Project Competition.

Technical Skills

- **AI & Computer Vision**: PyTorch, OpenCV, YOLO, ConvNeXt, Data Augmentation, GANs
- **Languages**: Python, C/C++, Java, Go, JavaScript, SQL
- **Web & Tools**: Vue.js, Gin (Go), Docker, Git, Linux (Ubuntu), LaTeX, CI/CD

Honors & Awards

- **Programming**: 2022 ICPC Asia Taiwan Online Programming Contest – Honorable Mention (Top 35%)